

Task-Dependent Variation. The random-update null model of Theorem 4.1 further implies that utility loss should be predictable from data overlap: tasks with distributions similar to $\mathcal{D}_i$ should cause degradation (overlapping data would induce updates with non-negligible projection onto M_i) while dissimilar tasks should be safe. However, [26] show that task influence is highly variable and poorly predicted by natural metrics. [25] formalize this by demonstrating that gradient-based selection finds harmful subsets within benign data—not by simple similarity-based computations.

All of these phenomena reveal a consistent signature for alignment degradation: it is systematic, it is rapid, and it is task-dependent (not similarity based).

4.3 Failure of Static Geometry

Our key observation is that the cause of the null model failing to explain observed empirical behaviors is its assumption that $F_i(\theta)$ is static, i.e., the sensitivity subspace M_i remains fixed throughout fine-tuning. Such an assumption is, in fact, only valid in a vanishingly small neighborhood of θ^* . In reality, both the magnitude and orientation of the Fisher information matrix evolve as parameters change along the trajectory $\theta(t)$. Thus, directions that were initially safe become dangerous. Moreover, fine-tuning updates themselves are highly structured in accordance with the task objective, and thus not random.

5 Alignment Instability Condition

Section 4 demonstrated that static, first-order analysis cannot explain empirical alignment degradation. The resolution requires understanding how fine-tuning trajectories evolve in a dynamic geometry where both update direction and the sensitive structure change over time.

We observe that in practice, fine-tuning induces a continuous trajectory $\theta(t)$ through parameter space along which the Fisher information varies: its spectrum shifts, its eigenspaces rotate, and the alignment sensitivity subspace M_i itself evolves. As a result, directions initially orthogonal to $M_i(\theta^*)$ need not remain, and small first-order projections can accumulate into significant degradation through higher-order effects. This section formalizes the precise structural conditions under which such accumulation becomes inevitable.

5.1 Intuition Behind Two Degradation Regimes

Let $g(\theta) = \nabla L_{\text{FT}}(\theta)$ denote the fine-tuning gradient and $M_i(\theta)$ the sensitivity subspace for skill S_i at parameters θ . We distinguish two fundamentally different mechanisms of alignment degradation based on the initial overlap (measured by Euclidean projection) between the update direction and sensitive subspace.

First-Order Regime. If the initial fine-tuning gradient has substantial projection onto M_i , that is, $\|P_i(\theta^*)g(\theta^*)\| \geq c$ for some non-negligible constant $c > 0$, then alignment degradation begins immediately. By the gradient flow $\dot{\theta}(t) = -g(\theta(t))$ with $\theta(0) = \theta^*$ we readily obtain that $\|P_i(\theta^*)(\theta(t) - \theta^*)\| = \Theta(ct)$. Combining with Proposition 3.3 yields quadratic degradation from first-order overlap. This regime corresponds to the more trivial fine-tuning tasks that directly modify alignment-relevant behaviors, requiring no additional geometric analysis beyond first-order projections. The remainder of this section focuses on the more subtle second-order regime.

Second-Order Regime. Consider instead the scenario where the initial gradient is nearly orthogonal to the sensitivity subspace: $\|P_i(\theta^*)g(\theta^*)\| < \varepsilon$ for small $\varepsilon > 0$. Here, the first-order term vanishes and, naively, the trajectory should remain safe. However, this conclusion assumes that M_i and g remain fixed. When accounting for how such quantities evolve, second-order terms dominate. Specifically, expanding $\theta(t)$ along the gradient flow to second order, we obtain:

$$\theta(t) - \theta^* = -tg(\theta^*) + \frac{t^2}{2}\nabla g(\theta^*)g(\theta^*) + O(t^3).$$

Under the projection, the first term contributes only εt , whereas the second term—the directional derivative of the gradient—can have substantial projection onto M_i . This acceleration term captures how the curvature of the fine-tuning loss bends the trajectory.

5.2 Geometric Intuition: Curvature as Steering

If parameter space were naturally Euclidean and $F_i(\theta)$ were constant, gradient flow would proceed in straight lines and a trajectory starting perpendicular to M_i would remain perpendicular, preserving alignment. In reality, the information geometry of neural networks is non-Euclidean [2, 54], with distance between distributions depending upon the position. Crucially, as the model adapts through parameter space, the metric changes, the subspaces rotate, and the gradient curves.

Curvature Coupling. The key quantity, $\nabla g(\theta^*)g(\theta^*)$, is the directional derivative of the fine-tuning gradient. Geometrically, this is the acceleration of the trajectory, or, how much the direction of steepest descent changes. Most important for the present study, even if $g(\theta^*)$ points away from M_i , the derivative $\nabla g(\theta^*)$ can point toward it. Through fine-tuning, we then have three interacting mechanisms dictating the misalignment process: M_i must have high curvature so movement into it damages safe behaviors (this determines loss magnitude, analogous to the first-order regime), the orthogonality of a trajectory to this subspace dictates potential movement within this space (initial condition of the second-order regime), and the non-trivial coupling of $\nabla g(\theta^*)g(\theta^*)$ can accelerate projection into this space (the acceleration mechanism unique to second-order regime).

5.3 Formal Definition

We now formalize the structural condition under which curvature forces trajectories into alignment-sensitive directions despite initial orthogonality. Throughout, for analytical clarity and without loss of generality, we assume that $F_i(\theta)$ and $g(\theta)$ are twice continuously differentiable in a neighborhood of θ^* .

Definition 5.1 (Alignment Instability Condition). *Fix a skill S_i . Let $F_i(\theta^*)$ have eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k \geq 0$, and let $M_i(\theta^*)$ be the span of the top d -eigenvectors of $F_i(\theta^*)$, with $P_i(\theta^*)$ the orthogonal projection onto $M_i(\theta^*)$. We say that θ^* satisfies the **Alignment Instability Condition (AIC)** for skill S_i with parameters $(d, \lambda, \gamma, \epsilon)$ if the following hold:*

1. **Low-Rank Sensitivity:** *The Fisher spectrum at θ^* is concentrated in a d -dimensional subspace: $\sum_{j>d} \lambda_j \leq \epsilon$ and $\lambda_d \geq \lambda$.*
2. **Initial Orthogonality:** *The fine-tuning update direction is initially nearly orthogonal to the sensitivity subspace: $\|P_i(\theta^*)g(\theta^*)\| \leq \epsilon$.*
3. **Curvature Coupling:** *The curvature of the fine-tuning objective induces nontrivial acceleration into the sensitivity subspace:*

$$\|F_i(\theta^*)^{1/2}P_i(\theta^*)\nabla g(\theta^*)g(\theta^*)\| \geq \gamma.$$

At a high level, the AIC identifies the regime where alignment appears stable initially (Condition 2) yet is structurally doomed (Condition 3) because it resides in a sharp, low-dimensional region (Condition 1). More formally, Condition 1 isolates the small set of directions where a skill S_i is fragile, with the bounds ensuring sufficient curvature concentrated in the first d eigenvectors and the tail contributing negligibly. Condition 2 captures the natural defense measure of projection away from sensitive subspaces, like that of [65]. However, we note that our analysis in Section 6 reveals this is a snapshot solution: unless projections are continuously updated to track the evolving sensitive subspace, curvature coupling (Condition 3) inevitably bends the trajectory back into sensitive regions over time. More specifically, this captures the rotational geometry crux: as we move along the trajectory, the gradient rotates toward M_i , with the parameter γ quantifying the strength of coupling.

6 Consequences of AIC

We now develop the main theoretical consequences of the AIC, establishing that alignment degradation under these conditions is not merely possible but structurally inevitable. These results enable both qualitative understanding (why benign tasks break safety) and quantitative prediction (how fast degradation occurs). Throughout this section, we fix a skill S_i and assume that θ^* satisfies Definition 5.1 with parameters $(d, \lambda, \gamma, \epsilon)$. Our approach proceeds in two steps, mirroring the geometric intuition from Section 5. We first quantify how much utility degrades when the trajectory enters the sensitive subspace M_i . Subsequently, we prove that the AIC forces the trajectory to enter M_i

at a quadratic rate. Combining the two results, we derive the quartic scaling law. The key insight is that these are separate geometric phenomena that when composed yield the quartic scaling law.

For clarity of presentation, the arguments in this section are carried out with respect to the sensitive subspace defined by the Fisher information at θ^* . We note, however, that the conclusions are robust to local variation of the Fisher information and to rotation of the sensitive subspace along the trajectory; accounting for these effects introduces only constant-factor and lower-order corrections and is deferred to Appendix B.

6.1 Projection of Misalignment Inevitability

We begin by formalizing the intuition that alignment resides in a small number of sharp directions. Movement into the subspaces M_i then degrades skill S_i , with loss growing quadratically in the projection magnitude.

Recall from Section 3 that utility loss locally takes the form $\Delta u_i \approx \frac{1}{2} \Delta \theta^\top F_i(\theta^*) \Delta \theta$. Here, the Fisher matrix characterizes the local curvature of alignment utility, identifying directions along which small parameter changes induce large distributional shifts. AIC Condition 1 concentrates the eigenspectrum, implying that movement in M_i causes this divergence to scale with λ whereas movement outside is negligible. Effectively, only the component projected into M_i matters for utility. The following theorem establishes the “cost” for entering sensitive regions.

Theorem 6.1. *Under Assumption 1 and assuming AIC holds, then there exists a neighborhood U of θ^* such that for all $\theta \in U$ with update step $\Delta \theta = \theta - \theta^*$:*

$$\Delta u_i(\theta) \geq \frac{1}{2} \|F_i^{1/2} P_i(\theta - \theta^*)\|^2 - O(\|\Delta \theta\|^3).$$

Consequently, if the projection satisfies $\|P_i(\Delta \theta)\| \geq \delta$ and the step size $\|\Delta \theta\|$ is sufficiently small, we have $\Delta u_i(\theta) = \Omega(\lambda \delta^2)$.

The proof follows from decomposing $F_i(\theta^*)$ and writing $\Delta \theta$ in the eigenbasis. Projecting this change with P_i then decomposes into the overlapping and orthogonal components, both of which can be bounded according to the AIC conditions. Full details are presented in Appendix A. Observe that this theorem characterizes the loss if projection occurs, but does not capture whether such overlap occurs during fine-tuning. We now focus on showing that, under the AIC, drift into M_i is dynamically forced.

6.2 Curvature-Induced Drift

Even when the initial fine-tuning gradient is nearly orthogonal to M_i , the curvature of the loss landscape forces the trajectory to drift into the sensitive subspace at a quadratic rate. This drift is dictated not by the velocity ($-g(\theta^*)$), but the acceleration ($\nabla g(\theta^*)g(\theta^*)$) which monitors how the direction of steepest descent changes along a trajectory. AIC Condition 3 bounds this factor, ensuring that projection M_i increases through training. Thus, our initially small overlap trajectory drifts into sensitive regions. Note that this acceleration is due to the curvature of the fine-tuning objective. Gradient descent follows the local geometry with no mechanism to anticipate or counteract curvature-induced drift.

Theorem 6.2. *Fix a skill S_i . Let $g(\theta) := \nabla L_{FT}(\theta)$ be the fine-tuning gradient and consider the gradient-flow trajectory defined by $\dot{\theta}(t) = -g(\theta(t))$ and $\theta(0) = \theta^*$. Assuming g is C^2 in an r -ball of θ^* and the AIC holds, then there exists a $t_0 > 0$ such that for $t \in (0, t_0]$ we have that*

$$\|F_i^{1/2} P_i(\theta(t) - \theta^*)\| \geq \frac{\gamma}{2} t^2 - \epsilon t - O(t^3).$$

The proof follows from Taylor expanding the trajectory along the gradient flow to second order. Projecting this trajectory onto M_i via P_i , we can then bound the resulting terms using the AIC in combination with the reverse triangle inequality to complete the result. Full details are in Appendix A.

6.3 Combing Geometry and Dynamics

Theorems 6.1 and 6.2 address complementary aspects of alignment degradation. Composing these results yields our main prediction: alignment utility degrades as the fourth power of training evolution.